

ZONAL'NOYE, Russian Federation

WMO: 299390

Lat: 52.6725N

Lon: 84.9289E

Elev: 739

StdP: 14.31

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -33.4	(c) -27.0	(d) -38.3	(e) 0.6	(f) -33.1	(g) -31.9	(h) 0.9	(i) -26.8	(j) 23.3	(k) 23.7	(l) 20.7	(m) 20.5	(n) 1.5	(o) 60	(p) 0.550

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.3	(c) 87.2	(d) 66.9	(e) 84.1	(f) 65.5	(g) 81.2	(h) 64.6	(i) 69.9	(j) 81.8	(k) 68.1	(l) 79.6	(m) 66.5	(n) 77.7	(o) 5.5	(p) 20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 66.0	(b) 98.4	(c) 74.9	(d) 64.0	(e) 91.7	(f) 73.1	(g) 62.3	(h) 86.3	(i) 71.9	(j) 34.4	(k) 82.2	(l) 32.9	(m) 79.9	(n) 31.7	(o) 77.9	(p) 81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	14.9	12.1	-39.2	93.1	8.3	3.7	-45.2	95.8	-50.0	97.9	-54.7	100.0	-60.7	102.6
			-39.1	73.5	7.7	3.7	-44.7	76.2	-49.2	78.4	-53.6	80.4	-59.2	83.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.1	1.3	6.2	21.7	40.5	54.4	64.7	68.0	63.9	52.7	39.3	21.6	8.3
	DBStd	25.77	15.09	14.38	12.14	10.71	9.31	7.02	5.75	6.59	7.84	9.29	14.05	14.23
	HDD50	6541	1509	1225	876	313	62	2	0	1	61	347	853	1292
	HDD65	10497	1974	1645	1341	735	345	91	36	100	375	796	1302	1757
	CDD50	1816	0	0	0	29	199	442	558	431	141	16	0	0
	CDD65	298	0	0	0	0	18	81	129	65	5	0	0	0
	CDH74	3151	0	0	0	23	289	808	1155	742	132	2	0	0
	CDH80	878	0	0	0	4	71	208	343	218	34	0	0	0
(14) Wind	WSAvg	4.7	4.3	4.4	4.9	5.7	5.4	4.4	3.8	3.7	4.0	5.1	5.3	5.1
(15) Precipitation	PrecAvg	21.0	1.3	1.1	0.9	1.4	2.0	2.1	2.3	2.5	1.8	2.0	1.9	1.6
	PrecMax	29.4	3.8	3.1	4.3	4.3	4.5	4.5	6.0	5.8	4.5	4.0	5.4	3.5
	PrecMin	13.0	0.2	0.1	0.0	0.2	0.3	0.5	0.2	0.7	0.5	0.4	0.2	0.2
	PrecStd	4.1	0.7	0.7	0.8	0.9	0.9	1.0	1.5	1.3	1.1	0.9	1.2	0.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.7	36.9	49.7	77.1	86.8	89.3	91.2	91.9	85.0	70.8	53.8	37.9
		MCWB	33.8	32.5	41.5	56.6	62.2	66.3	69.5	68.2	62.1	52.8	44.6	33.4
	2%	DB	31.9	33.2	44.5	69.4	81.5	85.1	87.1	85.5	77.2	63.9	47.5	34.2
		MCWB	29.6	30.0	38.0	52.8	59.6	64.9	68.5	66.7	59.4	50.3	41.1	30.9
	5%	DB	27.4	30.2	41.0	63.5	76.9	82.1	83.7	81.6	72.0	58.7	42.2	31.2
		MCWB	25.2	27.3	35.8	49.8	57.4	64.8	67.4	64.9	57.2	48.3	37.5	28.5
	10%	DB	22.7	26.5	37.4	57.8	71.9	78.8	80.7	77.9	67.4	53.7	37.6	27.8
		MCWB	20.9	24.0	33.4	46.9	55.7	63.3	66.5	63.6	55.0	45.7	34.3	25.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.0	33.2	42.6	57.3	65.3	69.8	73.8	71.3	64.2	54.3	45.9	34.2
		MCDB	36.5	35.5	48.9	73.6	81.9	82.4	85.3	85.4	78.3	67.1	51.2	36.7
	2%	WB	29.7	30.3	38.9	53.7	61.8	67.9	70.9	68.3	60.9	51.3	41.0	31.8
		MCDB	31.8	33.1	43.6	66.1	76.3	79.8	81.9	81.4	73.9	61.5	46.5	34.0
	5%	WB	25.4	27.3	36.2	50.5	59.5	66.2	69.0	66.4	58.6	49.0	37.9	28.5
		MCDB	27.5	29.9	40.3	62.1	72.4	78.2	79.9	78.8	70.0	57.9	42.0	30.6
	10%	WB	21.0	24.0	33.7	47.8	57.2	64.7	67.4	64.7	56.2	46.2	34.5	25.7
		MCDB	22.8	26.4	37.2	57.1	68.9	76.2	78.0	75.7	65.6	53.0	37.4	28.1
(35) Mean Daily Temperature Range	5% DB	MDBR	18.4	20.8	19.9	19.9	21.7	20.2	19.3	21.1	21.4	17.4	14.3	16.3
		MCDBR	16.9	19.0	19.2	30.6	29.4	25.9	24.5	27.6	30.6	28.2	17.6	16.2
		MCWBR	14.9	16.3	13.9	16.3	12.8	10.9	10.0	11.5	14.6	15.8	12.1	14.2
	5% WB	MCDBR	17.1	18.7	17.9	27.8	24.1	21.8	21.0	23.5	26.8	26.0	16.7	16.6
		MCWBR	15.5	16.4	13.6	15.7	11.9	10.3	9.8	10.7	13.8	15.3	12.0	15.2
(40) Clear-Sky Solar Irradiance	taub		0.299	0.338	0.391	0.424	0.416	0.412	0.436	0.412	0.373	0.363	0.346	0.310
	taud		2.011	1.944	1.906	2.076	2.176	2.272	2.212	2.255	2.310	2.213	2.033	1.957
	Ebn at Noon		202	230	243	255	264	267	257	257	253	224	180	168
	Edh at Noon		25	37	48	46	44	40	42	38	32	29	25	23
(44) All-Sky Solar Radiation	RadAvg		366	706	1165	1477	1767	2002	1821	1563	1112	599	324	256
	RadStd		32	40	111	97	180	132	117	102	101	69	40	30

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page